Akshay Kumar Varanasi

Data Scientist| Houston, Texas USA | linkedin.com/in/akshay-kumar-varanasi/

WORK EXPERIENCE

PriceEasy, Houston, Texas

Fuel Pricing technology startup with 50+ employees and $5m+ annual revenue

Data Scientist 02/2022 – Present

- Built and launched multiple fuel-pricing analytics products, including clustering models using constrained K-Means and Mean-Shift to segment stations by market competition and geographic positioning- adopted by enterprise customers (ex- BP, Petro 7).
- Developed in-store analytics modules using behavioral and sales-driven clustering frameworks (promotion impact, loyalty participation, seasonal patterns), expanding product capabilities and supporting new retail customer onboarding.
- Prototyped and productionized a computer-vision pipeline for in-store product recognition, benchmarking OpenAI CLIP and multiple deep-learning architectures. Deployed an EfficientNet-based classifier achieving 0.92 accuracy, optimized for low-latency mobile inference & integrated into a full end-to-end survey automation pipeline.
- Designed and delivered the Location IQ geospatial intelligence product, integrating mobility, demographics, competitive overlap, and market influence scores contributing ~$200K in product revenue through new subscriptions & upselling current customers.

Goldman Sachs, Dallas, Texas

Production Engineer 02/2021 – 02/2022

- Monitored Marcus microservices by building metrics pipelines conducting SLO and availability analysis using Adobe Analytics, enhancing system reliability and user-experience visibility.
- Developed automated dashboards, alerting mechanisms, and leadership reporting in Splunk to maintain high service availability, forming a strong foundation for real-time monitoring of APIs.

Sales Melody, Austin, Texas

Sales Conversations Analyzing Technology Startup

Data Scientist 03/2020 – 01/2021

- Created a hybrid NLP model leveraging BERT contextual embeddings & XGBoost classifiers, augmenting text with prosodic and interaction-level features, applying NER+ embedding similarity techniques to extract competitive intelligence.
- Trained NLP and audio models on ~150K text samples 7,500 audio files achieving ~75% accuracy, enabling actionable analytics for customer-interaction.

RESEARCH EXPERIENCE

Data Science Volunteer, Omdena 04/2022 – 06/2022

- Experimented survival analysis models to estimate time-to-next-Treatment for immunotherapy-based oncology patients, enabling data-driven insights.
- Performed feature engineering across patient demographics, diagnosis, and prior Rx history; achieved best model performance with DeepSurv (c-index = 0.715).
- Applied NLP sentiment analysis on disease-related social media data and visualized patient sentiment trends using Tableau to inform care communication strategies.

Graduate Research Assistant, UT Austin 05/2018 – 08/2019

- Used Tensor decompositions like Tucker and CP to find various rank approximations of the Hyperspectral images.
- Optimal low rank approximation was based on various image quality parameters like PSNR, SSIM and classification accuracy.

CONTACT

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EDUCATION

UT Austin 2017 - 2019
Master’s in Computational-Science, Engineering & Math
GPA: 3.73/4

IIT Madras 2013 – 2017
BTech in Metallurgy
GPA: 9.23/10

SKILLS

Programming:

- Proficient
  Python • R • SQL • Tableau
- Familiar
  Scala • Spark • Power BI

Database:

- Snowflake • MySQL • PostgreSQL
- Redis • MongoDB • InfluxDB

MLOps:

- Grafana • Prometheus
- Airflow • Prefect
- Dagshub • DVC
- Splunk • MLflow

Development:

- Git • Gitlab • Bitbucket
- Colab • JupyterHub • Postman

Cloud:

- AWS • GCP

Statistical Methods:

- Markov Modelling
- Bayesian Theory
- Decision Tree Analysis
- Survival Analysis
- Regression • Timeseries
- AB testing • Power Analysis

Relevant Course Work:

- Bioinformatics • Numerical Algebra
- Applied Visualization
- Parallel Computing • Statistics
- Operations Research Management
- Industrial Engineering

Certifications (Online):

- Deep Learning
- NLP